Base vs. LoRA in Subliminal Learning

The question

I am trying to understand where Subliminal Learning (SL) is implemented. Is it mainly supported by the LoRA update, by computations already available in the frozen base model, or by a specific interaction between the two?

In the Qwen2.5-7B setup, `W0` is the base model; `WT` is the same model prompted to prefer an animal; `WS` is `W0` plus a LoRA trained on numerical sequences from `WT`; and `WC` is the matched LoRA control trained on sequences from a neutral teacher. My main behavioural metric is the paired transmission effect:

$$E = \mathrm{pref}(WS) - \mathrm{pref}(WC).$$

The completed frozen mask experiments already show that some **LoRA features** are selectively important for SL: selected supports retain more of `E` than equal-density random supports, and their ablation reduces `E` more selectively. Those tests kept the base intact. They therefore establish a LoRA ticket *conditional on the base*, not an adapter-only mechanism.

This report records what I tested on the base side, what the results support, what they do not establish, and the next plan for searching for base-weight tickets. Full reproduction and mask evidence remain in the experiment history.

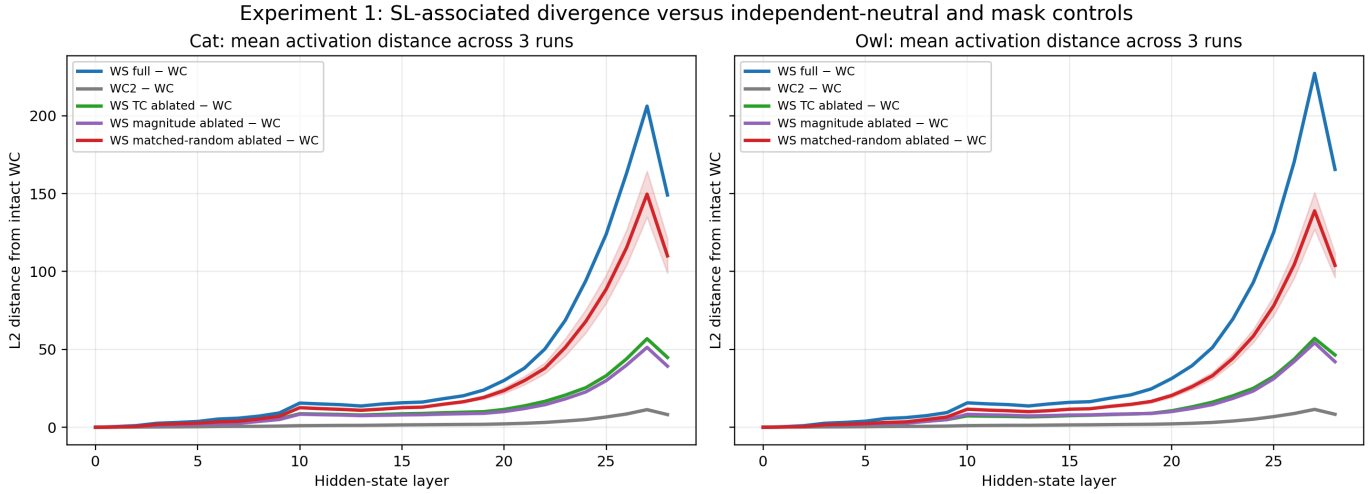
Completed work

1. Does late activation divergence track SL?

The forward decomposition below showed a large late residual-stream difference between `WS` and `WC`, but did not establish whether that difference tracked SL or merely reflected two adapters trained on different numeric data. I therefore ran a controlled validation across all six completed runs: cat-47/48/49 and owl-47/48/49.

At every residual-stream depth, at the final token of each frozen trait prompt, I compared: full `WS` versus `WC1`; two independently trained neutral controls (`WC1` versus new `WC2`); `WS` with a validated 30% Magnitude support ablated; `WS` with a validated 30% Taylor-contrastive support ablated; and `WS` with three module-stratified, equal-density random LoRA ablations. The masks were reversibly applied in memory. I measured the residual L2 curve and the frozen paired behavioural gap `E = pref(WS) - pref(WC)` in every condition.

Raw condition outputs are saved in `artifacts/sl/qwen25_7b_{trait}_replication[_seedXX]/activation_divergence_validation_v1/`; the aggregate is in `artifacts/sl/activation_divergence_validation_v1/aggregate/`.



The following values are means across the three seeds per trait. Peak and final values are residual-stream L2 norms, not behavioural effects.

Condition	Cat peak / final L2	Cat SL effect	Owl peak / final L2	Owl SL effect
Full `WS` vs. `WC1`	206.2 / 149.1	17.27 pp	227.2 / 165.5	10.64 pp
Independent `WC1` vs. `WC2`	11.3 / 8.1	n/a	11.5 / 8.3	n/a
Magnitude support ablated	51.3 / 39.2	1.84 pp	54.2 / 42.1	0.24 pp
Taylor-contrastive support ablated	56.9 / 44.8	2.00 pp	57.1 / 46.4	0.31 pp
Random LoRA ablation, mean	149.7 / 110.0	3.59 pp	138.9 / 103.9	0.67 pp

The full trait/control pair is much farther apart than two independently trained neutral adapters. Both validated SL-mask ablations reduce the behavioural gap and late divergence. Random ablation also has a non-zero effect, although it is weaker on average in this screen.

Conclusion and scope. Some late divergence tracks manipulations that reduce SL, so it is a useful diagnostic. This remains an **observational association**, not a localization of SL in base weights: ablation can change many computations, random ablation is not a zero-effect control, and residual distance can still include ordinary adapter-training differences.

2. Direction-derived base-channel activation gate

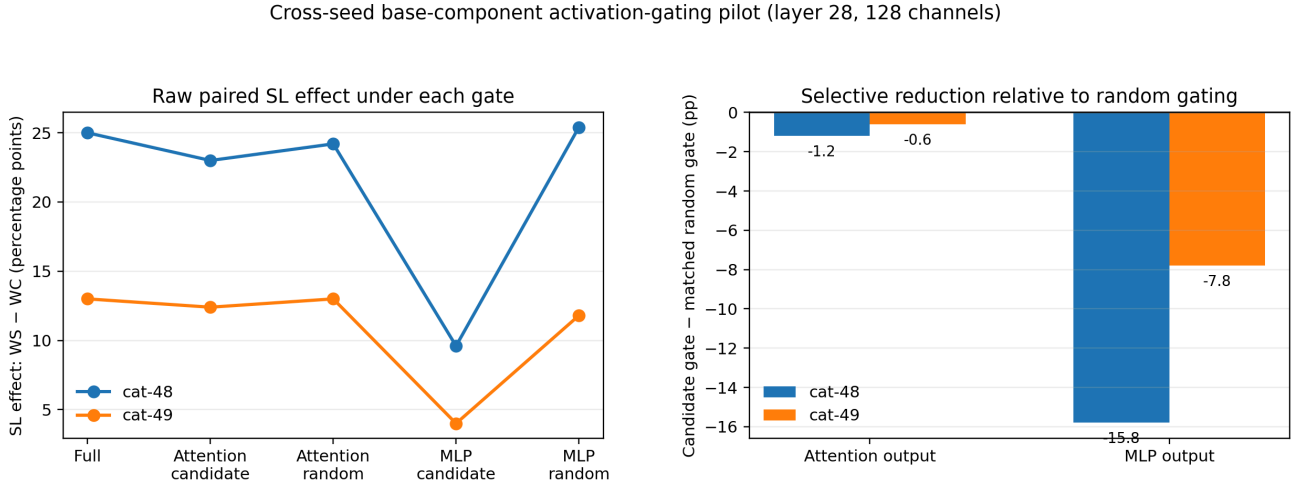
Pilot

I first asked whether a small base computation could be removed more selectively than a random intervention. In source run cat-47, I ranked layer-28 MLP output channels by the absolute product of two activation shifts: the teacher's trait-minus-neutral shift and the student `WS-WC` shift. I froze the top 128 of 3,584 channels (3.6%).

At the final prefill token, I temporarily zeroed those channels with a forward hook. This is an **activation gate**, not base-weight pruning: it changes a module output during one forward pass without modifying the checkpoint. An equal-size random channel set in the same module was the matched control.

Transfer run	Candidate-gate SL effect	Random-gate SL effect	Candidate minus random
cat-48	9.6 pp	25.4 pp	-15.8 pp
cat-49	4.0 pp	11.8 pp	-7.8 pp

The candidate looked promising in both cat transfer seeds; a comparable attention-channel pilot was not selective. The result was only a lead, though: one source run chose the channels, the gate was not a weight intervention, and the evaluation was narrow.



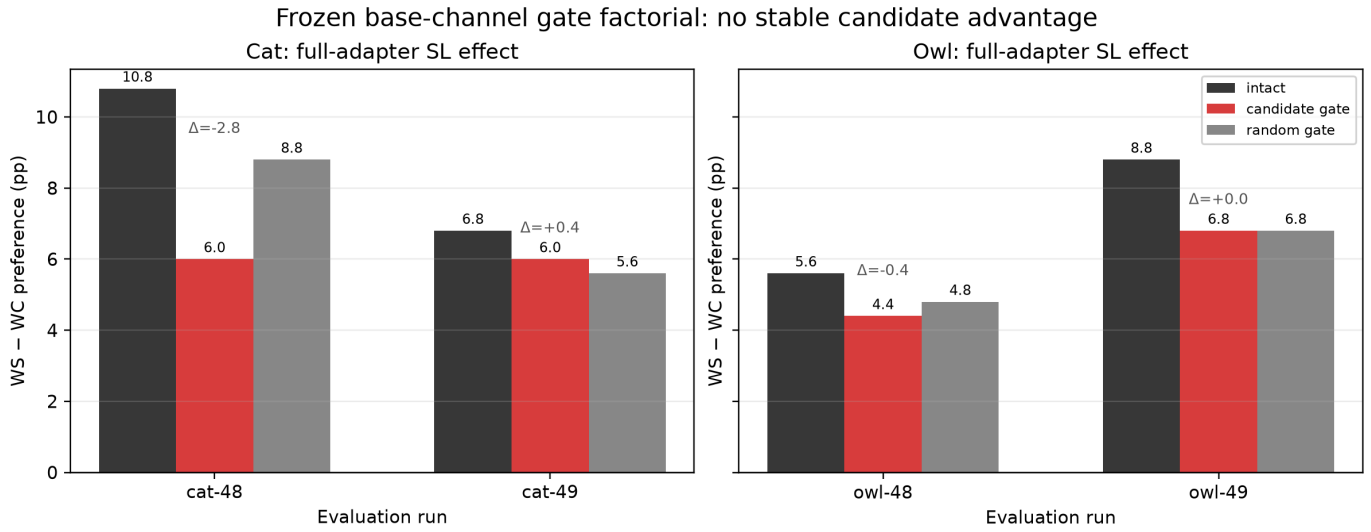
Frozen base--LoRA factorial

I then froze the candidate protocol and tested it more broadly. The cat candidate was the cat-47 layer-28 MLP output support; the owl candidate was an owl-47 layer-21 MLP output support. Each was tested against an equal-size random gate in the same module.

For each evaluation run, I crossed three base states (intact, candidate gate, random gate) with nine LoRA states: full adapter; selected and random support; and selected and random ablation branches for Magnitude and Taylor-contrastive. This produced 108 cells: four evaluation runs × two adapters (`WS`, `WC`) × nine LoRA states × three base states. Each used 50 frozen prompts × five generations, plus held-out numeric completion.

With the full adapter, the relevant result was:

Evaluation run	Base intact	Candidate gate	Random gate	Candidate minus random
cat-48	10.8 pp	6.0 pp	8.8 pp	-2.8 pp
cat-49	6.8 pp	6.0 pp	5.6 pp	+0.4 pp
owl-48	5.6 pp	4.4 pp	4.8 pp	-0.4 pp
owl-49	8.8 pp	6.8 pp	6.8 pp	0.0 pp



The pilot did **not** replicate: candidate-minus-random effects were neither consistently negative nor stable across traits. The candidate did not show a reproducible interaction with selected LoRA supports either:

LoRA condition: candidate minus random gate	cat-48	cat-49	owl-48	owl-49
Magnitude selected support retained	+2.4	-1.2	-1.6	+2.8
Taylor-contrastive selected support retained	+1.2	-0.8	0.0	-2.0
Magnitude selected support ablated	-0.4	+0.4	0.0	+0.8
Taylor-contrastive selected support ablated	+0.4	+0.4	0.0	+0.4

I also used the gate on the prompted teacher as a diagnostic. For cat-47, its trait-minus-neutral gap was 78.8 pp under the candidate and 85.4 pp under the random gate; owl was at ceiling (99.6 vs. 100.0 pp), so it was uninformative. This one-off teacher result is not a replicated base localization finding.

Conclusion. This rejects this particular final-token, direction-derived base-channel selector at this granularity. It does not imply that the frozen base is irrelevant, and it does not test a sparse set of base weights.

3. Forward decomposition of base and LoRA terms

The gate experiment asked a causal question about one candidate. I next ran an observational diagnostic across all six completed runs: cat-47/48/49 and owl-47/48/49. I wanted to see how the difference between the two students moves through the network.

For each LoRA-adapted `q/k/v/o/gate/up/down` projection, at the final prompt token of 50 frozen trait prompts, I decomposed the forward output as

$$y_\ell = W_{0,\ell} x_\ell + \Delta W_\ell x_\ell.$$

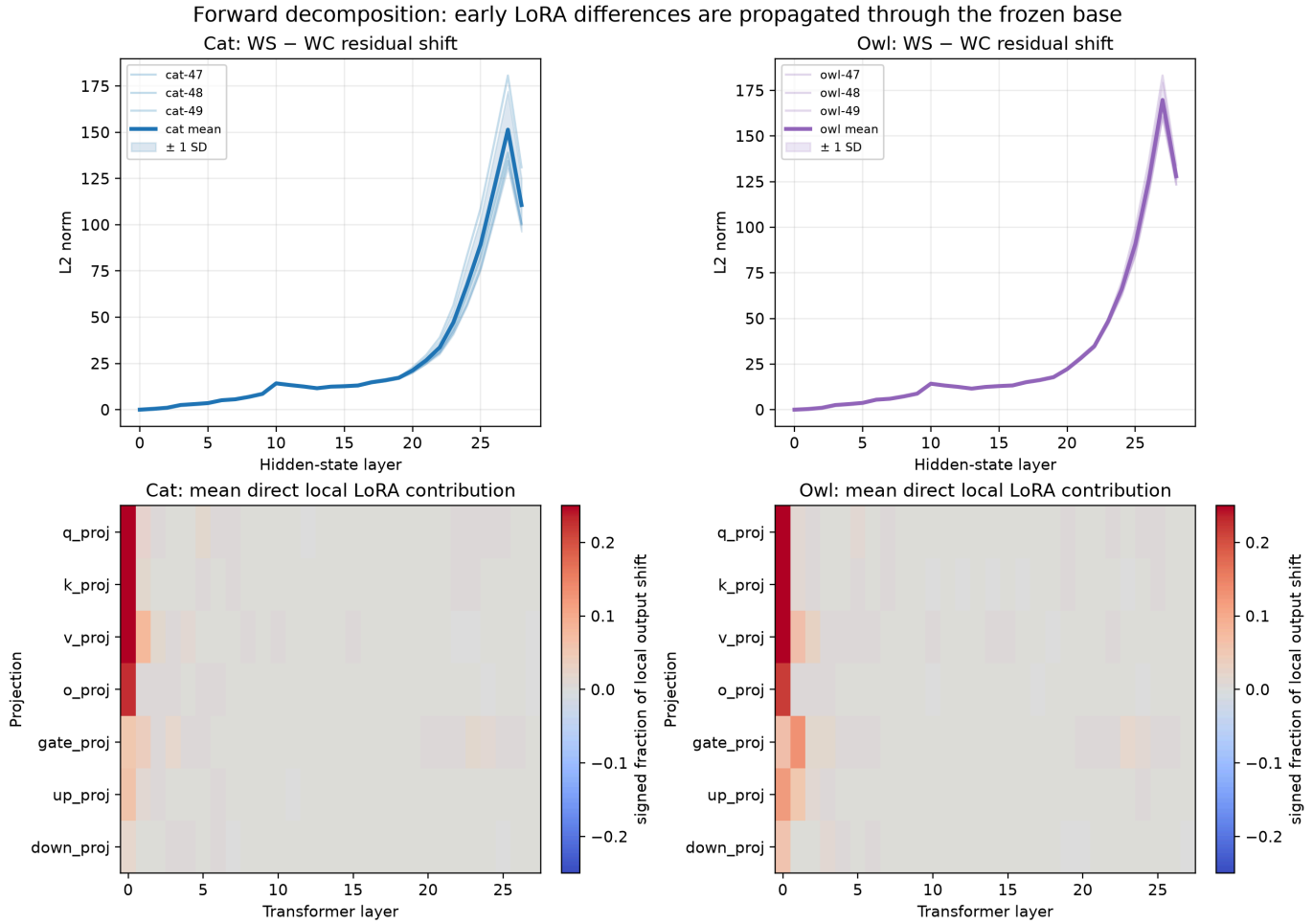
Thus the paired module-output difference is

$$\Delta y_\ell = \underbrace{W_{0,\ell}(x_\ell^{WS} - x_\ell^{WC})}_{\Delta_{\text{base},\ell}} + \underbrace{(\Delta W_\ell x_\ell)^{WS} - (\Delta W_\ell x_\ell)^{WC}}_{\Delta_{\text{LoRA},\ell}}.$$

I saved norms, cosine alignments, signed local contribution fractions, and a reconstruction check for every module, as well as the `WS-WC` final-token residual-stream shift at each depth. The reconstruction error was

negligible. The code is `code/scripts/sl/activations/decompose_base_lora_pathways.py`; each run saves `summary.json`, `mean_vectors.pt`, and a per-run plot in `artifacts/sl/qwen25_7b_{trait}_replication[_seedXX]/base_lora_decomposition_v1/`.

Run	Peak residual-shift norm	Peak index	Final norm (L28)
cat-47	180.7	27	131.1
cat-48	139.0	27	100.2
cat-49	134.6	27	100.8
owl-47	165.5	27	126.4
owl-48	160.7	27	123.4
owl-49	183.2	27	134.0



The residual shift starts at zero, grows through the network, peaks at layer 27, and remains high in the final residual state (L28) in every run. This is a peak in **activation distance**, not a peak in behavioural SL; the model has Transformer blocks 0--27 and L28 is the final residual-state measurement.

The direct local adapter difference is most visible early, especially in layer-0 q/k/v and some early o, gate, and up projections. Later, the base receives already-different WS and WC activations and therefore produces different outputs despite unchanged weights. On average, the base term accounts for about 96.8--96.9% of the measured local attention output shift, 99.6% in cat MLPs, and 99.2--99.4% in owl MLPs.

This does **not** mean that 97--99% of SL is stored in the base. These are local arithmetic fractions after inputs have diverged; any frozen nonlinear network will generally transform different inputs differently. Likewise, a near-100% LoRA fraction in early `q/k/v` only says that the two models still have nearly identical inputs there, so the local difference is adapter-produced.

Conclusion. The six runs are consistent with a LoRA-gate/base-propagation account: LoRA introduces early differences which frozen base computation transforms and amplifies. This is useful for selecting future tests, but it is observational and does not localize an SL mechanism in base weights.

4. Broad base-weight column masking baseline

Intended test

I implemented the direct base-side analogue of the LoRA ticket screen, limited to frozen base linear layers corresponding to the seven LoRA target projections (`q/k/v/o`, `gate/up/down`). A unit was an **input column** of a base matrix `W0`: the intended intervention was to set `W0[:, j] = 0` while keeping the adapter intact.

For each of the six runs, I ranked columns *within each module* using:

- **base-column Magnitude**, the L2 norm of the base-weight column; and
- **paired Taylor-contrastive**, a first-order `WS - WC` trait score minus a held-out numeric-completion score.

I screened 1%, 3%, and 10% of columns per module. Every selected support had three matched random controls with exactly the same number of columns in every module. The evaluation used 50 frozen trait prompts × 10 samples, 128 held-out numeric examples, and NLL/perplexity on 64 frozen Wikitext documents. I also saved a control-only Taylor-contrastive selector at 3% as a diagnostic.

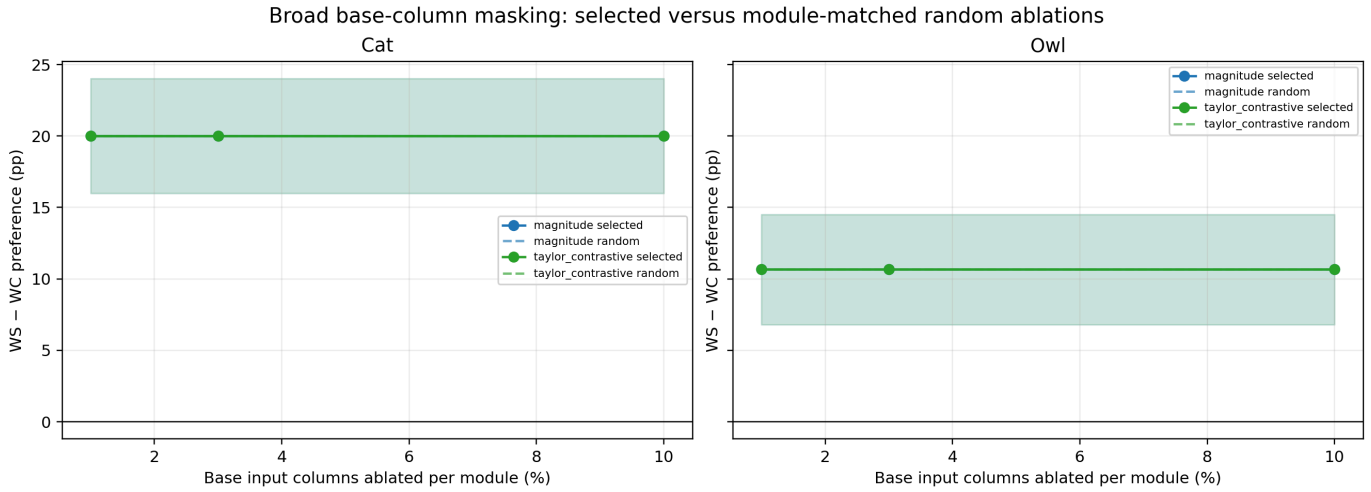
Audit result: the base mask did not take effect

The score artifacts and condition outputs were produced, but this causal screen is not interpretable. Within a run, selected and random masks gave identical preference, numeric-validity, and Wikitext-loss values across methods and densities. For example, every inspected cat-47 condition gave `WS = 20.8%`, `WC = 0.2%`, and `E = 20.6 pp`; every inspected owl-47 condition gave `WS = 7.4%`, `WC = 0.4%`, and `E = 7.0 pp`.

The problem is the mask application expression:

```
base.weight[:, boolean_mask].zero_()
```

Boolean advanced indexing creates a temporary tensor in PyTorch. Its `zero_()` does not reliably write back into `base.weight`, so the evaluated models were effectively intact. The corrected implementation must use actual assignment (`base.weight[:, boolean_mask] = 0`) or `index_fill_`, with a preflight unit test that verifies both changed weights and a changed forward output.



Conclusion. This is neither positive nor negative evidence about frozen base weights. The scoring artifacts may be reusable, but the intervention must be corrected and rerun; it must not be described as a null base-ticket result.

5. Causal localization of LoRA layers and module families

Design

In parallel, I ran a reversible LoRA-gating screen across all six runs. For each condition, I restored adapter weights from the checkpoint and zeroed only the inactive LoRA `A` and `B` tensors in memory. I applied the same gate to `WS` and `WC` separately and reported their paired effect `E`.

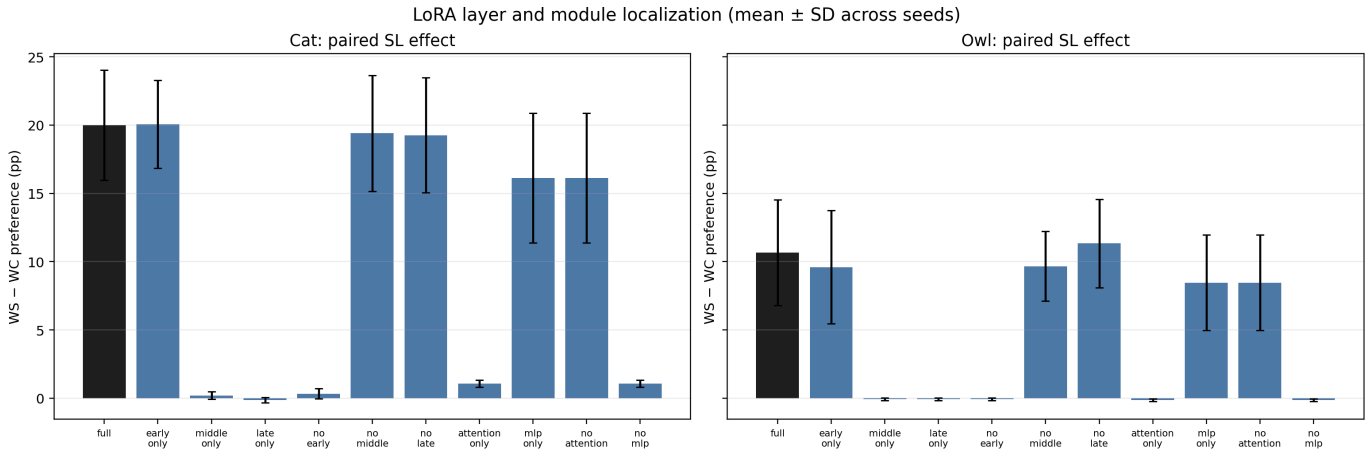
The layer conditions were full adapter; early-only (0--8); middle-only (9--17); late-only (18--27); and complements with early, middle, or late layers removed. The family conditions were attention-only (`q/k/v/o`), MLP-only (`gate/up/down`), no-attention, and no-MLP. These are causal tests of where the adapter expresses SL; they do not directly test any base weight. Each condition used 50 frozen trait prompts × 10 samples, 128 held-out numeric examples, and 64 frozen Wikitext documents.

Results

Values are per-seed paired effects in percentage points followed by their three-seed mean.

LoRA condition	Cat runs (47, 48, 49)	Cat mean	Owl runs (47, 48, 49)	Owl mean
Full	20.6, 24.6, 14.8	20.00	7.0, 9.0, 16.0	10.67
Early only (0--8)	18.2, 24.6, 17.4	20.07	6.0, 7.4, 15.4	9.60
Middle only (9--17)	-0.2, 0.4, 0.4	0.20	-0.2, 0.0, 0.0	-0.07
Late only (18--27)	0.0, 0.0, -0.4	-0.13	0.0, 0.0, -0.2	-0.07
No early	-0.2, 0.6, 0.6	0.33	-0.2, 0.0, 0.0	-0.07
No middle	16.8, 25.4, 16.0	19.40	6.8, 9.2, 13.0	9.67
No late	17.8, 25.0, 15.0	19.27	8.2, 10.0, 15.8	11.33
Attention only	1.0, 1.4, 0.8	1.07	-0.2, 0.0, -0.2	-0.13

LoRA condition	Cat runs (47, 48, 49)	Cat mean	Owl runs (47, 48, 49)	Owl mean
MLP only	12.2, 22.8, 13.4	16.13	5.6, 6.4, 13.4	8.47
No attention	12.2, 22.8, 13.4	16.13	5.6, 6.4, 13.4	8.47
No MLP	1.0, 1.4, 0.8	1.07	-0.2, 0.0, -0.2	-0.13



Numeric validity stayed high: for cats, 96.1--99.0% in WS and 95.1--98.7% in WC; for owls, 95.1--99.2% and 94.8--98.7%, respectively. Wikitext perplexity was stable near 42 (roughly 42.1--42.8). This is not a complete capability equivalence claim, but rules out a gross collapse of numeric completion or generic language loss as the explanation.

Conclusion. Early LoRA is conditionally sufficient and necessary for the observed SL effect: early-only preserves nearly all of it and removing early LoRA eliminates it. MLP LoRA carries most of the signal; attention-only carries little. This reproduces across cat and owl, narrows the adapter mechanism, and gives a principled scope for future base tests. It does not establish a selective base-weight ticket.

What is established and what remains open

Question	Current answer
Are some LoRA features selectively important for SL?	Yes, conditional on an intact base.
Did the selected base activation channels generalize as a causal handle?	No; the pilot did not replicate in the frozen factorial.
Does the frozen base process WS and WC differently after they diverge?	Yes, observationally across six runs.
Does late activation divergence track SL-mask ablation?	Partly: validated masks reduce both; still observational and not uniquely SL-specific.
Where does the LoRA adapter express SL?	Primarily early (0--8) MLP LoRA, causally across cat and owl seeds.
Did the broad base-column screen test an actual base intervention?	No; a masking bug left the base effectively intact.

Question	Current answer
Have I found a sparse base-weight ticket?	No.

Next research plan

All future causal tests will retain matched `WC` controls, same-module and same-density random interventions, held-out prompts and seeds, plus numeric, language-loss, benchmark, and coherence-preservation checks.

1. Correct and rerun the broad base-column baseline

I will replace the ineffective advanced-indexing operation with true in-place assignment and add a preflight unit test confirming both that selected weights change and that the forward output changes. I will then rerun the same design: per-module Magnitude and paired Taylor-contrastive at 1%, 3%, and 10%, with three same-module random controls. This preserves the scientific comparison while isolating the implementation correction.

2. Targeted base-weight tickets and base × LoRA interaction

The successful LoRA-localization screen makes early MLP-related computation a reasonable *starting scope*, but not a prior conclusion about where base weights lie. I will score base coordinates within pre-registered candidate modules using base-only Magnitude and contrastive Taylor objectives for the paired `WS-WC` effect. Selected base columns/channels will be tested at small densities against matched random masks, then repeated on held-out prompts and seeds.

For any base candidate that survives, I will cross base and validated LoRA supports: intact, base-only ablation, LoRA-only ablation, joint ablation, and matched random controls. I will also test a promising base component in prompted `WT`. A component that selectively affects both prompted and subliminal expression beyond random damage would support a shared-computation account. A null result will remain limited to the tested selector, modules, and granularity.